

$f: M \rightarrow M \rightarrow \text{surjection (injection)} \rightarrow f \text{ is iso}$
 $\uparrow$
 Noetherian (Artinian)

$$M/\ker f \cong M$$

$$[Ler T^n \cap Im T^n]$$

$$|L| \leq T^n$$

$$0 \xrightarrow{f} L \xrightarrow{g} M \xrightarrow{g} N \rightarrow 0$$

$$M : \mathcal{H}(E) \hookrightarrow L_\infty N : \mathcal{H}(E)$$

$$(Art', n) \quad (Art', n)$$

$$M/f(L) \cong N$$

$$\mathbb{M} \cong \mathbb{N}$$

Artinian ring

$$d \subset \subset$$

$$\mathbb{Z} \subset \mathbb{Z}^2 \subset \mathbb{Z}^3 \subset \dots$$

$$0 \rightarrow L \xrightarrow{f} M \hookrightarrow f^* \mathcal{I}_Y$$

$$M \xrightarrow{g} N \rightarrow 0 \hookrightarrow g^* \mathcal{I}_X$$

(Cor) $M_i \in \mathcal{L}_i \quad (1 \leq i \leq n)$
 $\Rightarrow \bigoplus_{i=1}^n M_i \in \mathcal{L}_i$

$$0 \rightarrow M_1 \rightarrow \bigoplus_{i=1}^n M_i \rightarrow \bigoplus_{i=1}^{n-1} M_i \rightarrow 0$$

$T: V \rightarrow V$ is diagonalizable $\Leftrightarrow m_T$ splits and separable

" triangulizable $\Leftrightarrow m_T$ is separable

E/F is separable $\Leftrightarrow \# \text{Emb}_F(E, \bar{F}) = [E:F]$

E/F is splitting $\Leftrightarrow \text{Emb}_F(E, \bar{F}) = \text{Aut}(E/F)$

relation $\{ \}$

$$\alpha, \beta \in \mathbb{R}^3, \quad r \in \mathbb{R}^3,$$

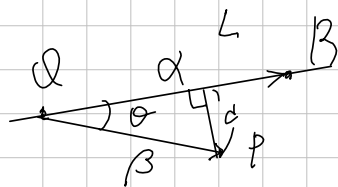
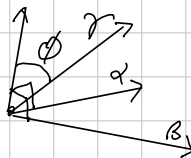
$$1. |\alpha \times \beta| = |\alpha| \cdot |\beta| \cdot \sin \theta.$$

$$2. \alpha \cdot \beta = |\alpha| \cdot |\beta| \cdot \cos \theta.$$

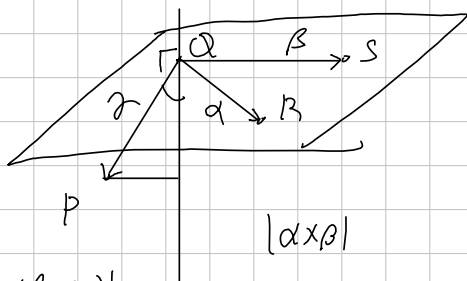
$$3. V = A \cdot h = \underbrace{|\alpha| \cdot |\beta| \cdot \sin \theta}_{|\alpha \times \beta|} \cdot \underbrace{h}_{|r| \cdot \cos \phi}$$

$$= |\alpha \times \beta| \cdot |r| \cdot \cos \phi$$

$$= |r \cdot (\alpha \times \beta)|.$$



$$d = \frac{|\alpha \times \beta|}{|\alpha|}$$



$$d = \frac{|\alpha \cdot (\beta \times r)|}{|\alpha \times \beta|}$$

$$L: \mathbb{R} \rightarrow \mathbb{R}^3: t \mapsto (a \ b \ c) \cdot (t) = \begin{pmatrix} at \\ bt \\ ct \end{pmatrix}$$

$$T: \mathbb{R}^3 \rightarrow \mathbb{R}: (x, y, z) \mapsto (a \ b \ c) \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$n = (a, b, c), \text{ and } P_0 = (x_0, y_0, z_0).$$

$$n \cdot (P_0 - P) = 0 \Leftrightarrow a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.$$

1). $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$ does not exist.

Proof. Take $f(x,y) = \frac{x^2 - y^2}{x^2 + y^2}$.

First, observe that $\lim_{y \rightarrow 0} f(0,y) = -1$

but, $\lim_{x \rightarrow 0} f(x,0) = 1$. So that's proved.

2). $f(x,y) = \frac{xy}{x^2 + y^2}$

Observe that $\lim_{x \rightarrow 0} f(x,x) = \frac{1}{2}$

but $\lim_{x \rightarrow 0} f(x,0) = 0$ $a+b \geq 2\sqrt{ab}$

3). $f(x,y) = \frac{x^2 y}{x^2 + y^2}$

Given $\epsilon > 0$, take $\delta = \epsilon$. Then, $\|(x,y)\| < \delta \Rightarrow$

$$|f(x,y) - 0| = \left| \frac{x^2 y}{x^2 + y^2} \right| \leq \frac{|x^2 y|}{2|x y|} = \frac{1}{2} |x| < \delta = \epsilon.$$

$\{X^1, X^2, \dots, X^p\}$ is linearly indep-

$\Leftrightarrow$

$$\sum_{k=1}^p a_k \cdot X^k = 0 \Rightarrow a_1 = a_2 = \dots = a_p = 0$$

$$\begin{pmatrix} X^1 & \dots & X^p \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_p \end{pmatrix} = u_1 X^1 + \dots + u_p X^p$$

assume $a_1 X^1 + \dots + a_q X^q = 0$. since

$$= a_1 X^1 + \dots + a_q X^q + 0 \cdot X^{q+1} + \dots + 0 \cdot X^p$$

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$X \mu$

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$X \mu = 0$ 일 때 $\mu = 0$ 으로 유도하라

$$(\mu_1 + \mu_3)X + (\mu_1 + \mu_2)Y + (\mu_2 + \mu_3)Z = 0$$

$$\begin{array}{l} 1+1=0 \\ 1+1=0 \\ 2=0 \end{array}$$

$$\begin{cases} \mu_1 + \mu_3 = 0 \\ \mu_1 + \mu_2 = 0 \\ \mu_2 + \mu_3 = 0 \end{cases}$$

$$\mu_3 - \mu_2 = 0$$

$$2=0$$

$$2 \cdot \mu_3 = 0$$

$\mathbb{R} \quad \mathbb{C}$

$$\cdot \mu_3 = 0 \quad \frac{1}{2}$$

$f: \mathbb{R}^2 \rightarrow \mathbb{R}$ is differentiable at $(x, y) = (a, b)$

$$\Leftrightarrow \lim_{h \rightarrow 0} \frac{|f(a, b) + h| - f(a, b) - \mathbb{R} \cdot h|}{\|h\|} = 0$$

$$f = \frac{1}{2}x^2 + 4xy + h(y),$$

$$Z - Z_0 = f_x(x - x_0) + f_y(y - y_0)$$

$$f_y = 4x + h'(y)$$

EXAMPLE 3. Define $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ by setting $f(0) = 0$ and

$$f(x, y) = x^2 y / (x^4 + y^2) \quad \text{if } (x, y) \neq 0.$$

We show all directional derivatives of f exist at 0 , but that f is not differentiable at 0 . Let $u \neq 0$. Then

$$\begin{aligned} \frac{f(0 + tu) - f(0)}{t} &= \frac{(th)^2(tk)}{(th)^4 + (tk)^2} \frac{1}{t} \quad \text{if } u = \begin{bmatrix} h \\ k \end{bmatrix} \\ &= \frac{h^2 k}{t^2 h^4 + k^2}, \end{aligned}$$

so that

$$f'(0; u) = \begin{cases} h^2/k & \text{if } k \neq 0, \\ 0 & \text{if } k = 0. \end{cases}$$

$$\Delta Z = f(a + \Delta x, b + \Delta y) - f(a, b)$$

$$dZ = f_x(x, y) dx + f_y(x, y) dy$$